Conditioned Coherence Index for Predictive Beamforming Switching in Wi-Fi 7/6G Systems

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Abstract*— High-frequency wireless systems such as Wi-Fi 7 (IEEE 802.11be), millimeter-wave (mmWave), and emerging 6G networks operate under rapid channel variation, where Channel State Information (CSI) becomes quickly outdated. Unlike traditional systems that degrade gradually, these environments exhibit abrupt performance collapse under mobility. This paper introduces the Coherence Cliff, a boundary at which temporal channel decorrelation and spatial matrix instability jointly cause beamforming failure. We derive the Conditioned Coherence Index (CCI), a metric that combines Doppler-induced decorrelation with channel matrix conditioning to predict instability. We show that as CCI approaches a threshold, Zero-Forcing (ZF) becomes unstable due to noise amplification. To address this, we propose a predictive switching mechanism between ZF and Maximum Ratio Transmission (MRT), enabling stable operation under mobility. This framework provides a practical approach for improving robustness in next-generation wireless systems.

Keywords-Wi-Fi 7, 6G, MIMO, Beamforming, Channel State Information (CSI), Adaptive Precoding

I. INTRODUCTION

The increasing demand for high data rates and low latency has driven wireless systems toward higher frequencies, from the 6 GHz band used by Wi-Fi 7 (IEEE 802.11be) [10], [13] to millimeter-wave and terahertz bands targeted by emerging 6G technologies [6], [7]. While providing greater bandwidth, these frequencies introduce propagation, mobility, and channel-estimation challenges [9]. Under mobility, reduced coherence time causes CSI to become outdated more rapidly [8], degrading beamforming performance [1], [2], [11].

This degradation can become abrupt when temporal channel decorrelation is combined with spatial matrix ill-conditioning, which makes Zero-Forcing (ZF) precoding highly sensitive to perturbations [3]. We define this transition as the Coherence Cliff and introduce the Conditioned Coherence Index (CCI) to jointly capture temporal and spatial instability. Using CCI, we propose predictive switching between ZF and Maximum Ratio Transmission (MRT) to maintain beamforming stability under mobility.

The main contributions are Doppler-based channel decorrelation analysis, the Conditioned Coherence Index (CCI) for predicting beamforming instability, linking channel variation to precoding degradation, and predictive ZF-to-MRT switching before the Coherence Cliff.

II. SYSTEM MODEL

A. Channel Model

$$h(t) = \sum_{n=1}^{N} a_n e^{j2\pi f_n t}$$

(1)

where $h(t)$ represents the time-varying wireless channel response, a_n is the complex gain of the n-th multipath component, $f_n$is the Doppler frequency of the n-th path, $N$is the total number of multipath components, and $j = \sqrt{-1}$is the imaginary unit. This model represents multipath propagation with varying delays, amplitudes, and Doppler shifts. User motion causes faster phase variation, reducing channel correlation and CSI accuracy, particularly at high frequencies. Consequently, beamforming becomes less reliable and increasingly prone to instability.

B. MIMO System Model

$$y = HWs + n$$

(2)

where H is the channel matrix, W is the precoding matrix, **s** is the transmitted signal vector, and n is additive noise [4], [5]. Precoding performance depends on accurate channel estimation and matrix conditioning; when H is ill-conditioned, even small estimation errors can significantly degrade performance.

C. Precoding Schemes

Two canonical linear precoding strategies are considered.
The Zero-Forcing (ZF) precoding matrix is designed to minimize multi-user interference by inverting the channel:

$$W_{ZF} = H^H (HH^H)^{-1}$$

(3)

where H is the MIMO channel matrix. ZF suppresses inter-user interference through channel inversion but becomes sensitive to perturbations as H becomes ill-conditioned, amplifying noise and estimation errors. In contrast, MRT maximizes received signal power without channel inversion:

$$W_{MRT} = \alpha H^{\boldsymbol{H}}$$

(4)

even when H is ill-conditioned, at the cost of not suppressing inter-user interference [3], [12]. This trade-off — ZF's interference suppression versus MRT's robustness to ill-conditioning — is the basis for the predictive switching mechanism introduced in Section V.

D. Doppler and Coherence

$$f_D = \frac{v\, f_c}{c}$$

(5)

$$T_c \approx \frac{0.423}{f_D}$$

(6)

where f_D denotes the maximum Doppler frequency, v is the relative velocity between the transmitter and receiver, f_c is the carrier frequency, c is the speed of light, and T_c represents the channel coherence time. These expressions quantify the impact of mobility on channel variation. As user velocity increases, Doppler frequency increases, resulting in shorter coherence time. This means that the channel changes more rapidly, reducing the validity duration of CSI. This relationship is particularly significant in high-frequency systems, where even moderate mobility can lead to rapid CSI aging. As a result, beamforming decisions based on outdated CSI can become highly inaccurate.

E. Channel Autocorrelation

$$\rho\,(\tau) = J_0(2\pi\, f_D \tau)$$

(7)

This function characterizes how similar the channel remains over time delay τ. As τ increases, the correlation decreases, indicating that the channel is becoming less predictable. The Bessel function J_0 captures the statistical behavior of fading in mobile environments. The key insight here is that temporal decorrelation is not linear; rather, it follows a nonlinear pattern that can lead to rapid loss of predictability beyond a certain threshold.

F. Matrix Sensitivity

$$\frac{||\Delta W||}{||W||} \leq \kappa(H) \cdot \left(\frac{||\Delta H||}{||H||}\right)$$

(8)

where κ(H) is the condition number of the channel matrix. As shown in (8), the stability of the precoding operation is directly tied to the conditioning of the channel matrix. A large condition number implies that even small perturbations in H can lead to large deviations in W. This relationship is critical in understanding why high-frequency systems experience abrupt instability. As the channel evolves and becomes less structured, its condition number can increase rapidly, pushing the system toward instability.

III. THE COHERENCE CLIFF

The Coherence Cliff emerges from the interaction between temporal and spatial instabilities. As the channel decorrelates over time and the matrix becomes ill-conditioned, the system approaches a critical point, expressed as (9).

$$\sigma_{\min}(H) \rightarrow 0$$

(9)

The minimum singular value of the channel matrix approaching zero indicates that the matrix is becoming singular. In this regime, Zero-Forcing precoding becomes highly unstable and amplifies noise dramatically [3].
This transition is not gradual. Instead, it represents a sharp boundary where performance collapses. Identifying this boundary is crucial for predictive control.

IV. CONDITIONED COHERENCE INDEX (CCI)

We define the Conditioned Coherence Index (CCI) as a unified metric, given by (10).

$$CCI(t) = \frac{\kappa(H) - 1}{\kappa(H) - 1 + \kappa_0} \cdot [1 - |J_0(2\pi f_D \tau)|] \cdot \frac{\alpha}{SINR(t) + \alpha} \cdot [1 - e^{-\beta\tau/T_c}]$$

(10)

where κ(H) is the condition number of channel matrix H; $J_0(\cdot)$ is the zeroth-order Bessel function of the first kind; f_D is the maximum Doppler frequency; τ is the feedback delay; SINR(t) is the instantaneous signal-to-interference-plus-noise ratio; α is the signal sensitivity coefficient; β is the aging decay rate; and T_c is the coherence time.
For the simulations, the CCI parameters were set to $\kappa_0 = 10$, α = 10, and β = 1. The value α = 10 controls the sensitivity of the SINR penalty and places its transition around an SINR of 10 on the linear scale. The value β = 1 provides a normalized CSI-aging response, where the aging penalty increases as the CSI delay approaches the channel coherence time. These values were kept constant across all simulated scenarios to provide a consistent comparison.
The goal of the CCI equation is to combine four different properties of a wireless MIMO channel: 1) spatial channel conditioning, 2) time variation caused by motion, 3) signal quality, and 4) CSI aging.

The first component of the proposed Conditioned Coherence Index (CCI) quantifies spatial channel conditioning through the normalized condition-number term, $\frac{\kappa(H)-1}{\kappa(H)-1+\kappa_0}$, where $\kappa(H) = \sigma_{\max}(H)/\sigma_{\min}(H)$. The condition number measures the numerical stability of the MIMO channel matrix during precoding. When the channel is well conditioned, the singular values are comparable, the condition number approaches one, and the penalty remains close to zero. As user channels become increasingly correlated, the smallest singular value decreases, causing the condition number to rise rapidly. This increases the sensitivity of Zero-Forcing (ZF) precoding to channel estimation errors and noise amplification, making spatial instability a major contributor to beamforming degradation.

The second component represents time variation caused by user motion and is modeled by the penalty term $1-|J_0(2\pi f_D \tau)|$, where $J_0(\cdot)$ is the zeroth-order Bessel function describing temporal channel autocorrelation. The Doppler frequency f_D increases with user velocity and carrier frequency, causing the channel to decorrelate more rapidly. As either mobility or CSI feedback delay increases, the channel estimate becomes progressively less representative of the current propagation environment. Consequently, beamforming weights computed from outdated CSI become less accurate, increasing the likelihood of precoding instability. The penalty therefore increases as temporal correlation decreases, reflecting the loss of channel predictability.

The third component evaluates signal quality using the normalized penalty $\frac{\alpha}{\mathrm{SINR}(t)+\alpha}$. Unlike conventional reliability measures, this term is formulated so that larger values indicate greater instability. When the received Signal-to-Interference-plus-Noise Ratio (SINR) is high, interference and noise have minimal impact, keeping the penalty close to zero. As SINR decreases because of fading, interference, or noise, the penalty approaches one, indicating deteriorating communication conditions. The parameter α controls the sensitivity of the transition and determines how strongly poor signal quality contributes to the overall instability index.

The fourth component models CSI aging through the exponential penalty $1 - e^{-\beta\tau/T_c}$, where τ is the CSI delay, T_c is the channel coherence time, and β controls the aging rate. When the delay is much smaller than the coherence time, CSI remains accurate and the penalty is nearly zero. As the delay approaches or exceeds the coherence time, the channel evolves significantly before the CSI can be used, causing the penalty to increase toward one. This term explicitly captures the degradation caused by stale channel information and complements the temporal decorrelation term by measuring CSI freshness.

The CCI combines four normalized instability measures into a metric bounded between 0 and 1, where higher values indicate greater beamforming instability. Simulations identify CCI = 0.6 as the ZF-to-MRT switching threshold, where precoding error begins to increase rapidly. This enables predictive switching before the Coherence Cliff, improving robustness under rapidly varying channels.

V. PREDICTIVE BEAMFORMING SWITCHING

The optimal beamforming strategy is selected based on the Conditioned Coherence Index (CCI), which captures the combined effects of Doppler-induced temporal decorrelation, channel matrix conditioning, signal quality, and CSI aging. The switching rule is defined in (11).

$$W_{opt} = \begin{cases} \mathrm{W_{ZF}}, \mathrm{CCI(t)} < \gamma \\ \mathrm{W_{MRT}}, \mathrm{CCI(t)} \geq \gamma \end{cases} \quad (11)$$

where γ denotes the transition boundary between stable and unstable precoding regimes. Based on the CCI analysis in Section IV and the simulation results in Section VI, we adopt $\gamma = 0.6$ as the predictive switching threshold. Across the simulated mobility and CSI-aging conditions, the CCI consistently approaches 0.6 at the onset of rapid perturbation-error growth and noticeable degradation of Zero-Forcing (ZF) performance. This value therefore represents the earliest reliable transition point at which temporal channel decorrelation and channel matrix ill-conditioning begin to jointly compromise precoding stability, while ZF still provides effective interference suppression. Selecting $\gamma = 0.6$ enables proactive switching to Maximum Ratio Transmission (MRT) before the onset of the Coherence Cliff, balancing interference cancellation under favorable channel conditions with robustness under rapidly varying channels.

Existing methods mitigate mobility-induced degradation through fixed ZF/MRT schemes [14], learning-based channel prediction [15], or enhanced CSI feedback [16]. In contrast, the proposed Conditioned Coherence Index (CCI) provides a unified closed-form decision metric that integrates temporal decorrelation and channel matrix conditioning to enable lightweight predictive switching without requiring additional training, iterative optimization, or signaling overhead.

VI. SIMULATION RESULTS

Simulation Setup— Simulations considered an 8-antenna downlink serving four users at 6 GHz using a Jakes-based Rayleigh fading channel. User velocity varied from 0–50 m/s, with a 1-ms CSI feedback delay and SINR ranging from 0–30 dB. Results were evaluated over 1000 channel realizations. The CCI parameters were $\kappa_0 = 10$, $\alpha = 10$, and $\beta = 1$, with switching threshold $\gamma = 0.6$.

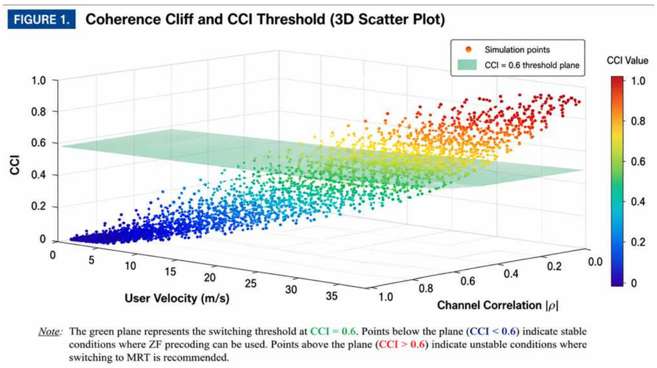

Fig. 1. Visualization of the Coherence Cliff and CCI-based switching threshold.

The three-dimensional scatter plot shows CCI as a function of channel correlation and user velocity. Blue points indicate stable, highly correlated channels (low CCI), whereas yellow points indicate decorrelated, unstable channels (high CCI). The plane at CCI = 0.6 marks the proposed ZF-to-MRT switching threshold.

The threshold $\gamma = 0.6$ was selected based on a sensitivity trade-off between Zero-Forcing (ZF) error growth and Maximum Ratio Transmission (MRT) stability. A lower threshold (e.g., $\gamma = 0.5$) causes premature switching to MRT, needlessly sacrificing multi-user interference cancellation under manageable mobility. Conversely, a higher threshold (e.g., $\gamma = 0.7$) delays switching past the Coherence Cliff, exposing ZF to exponential error growth (exceeding 10^1 relative error). Therefore, $\gamma = 0.6$ serves as the exact numerical tipping point where MRT outperforms degraded ZF precoding

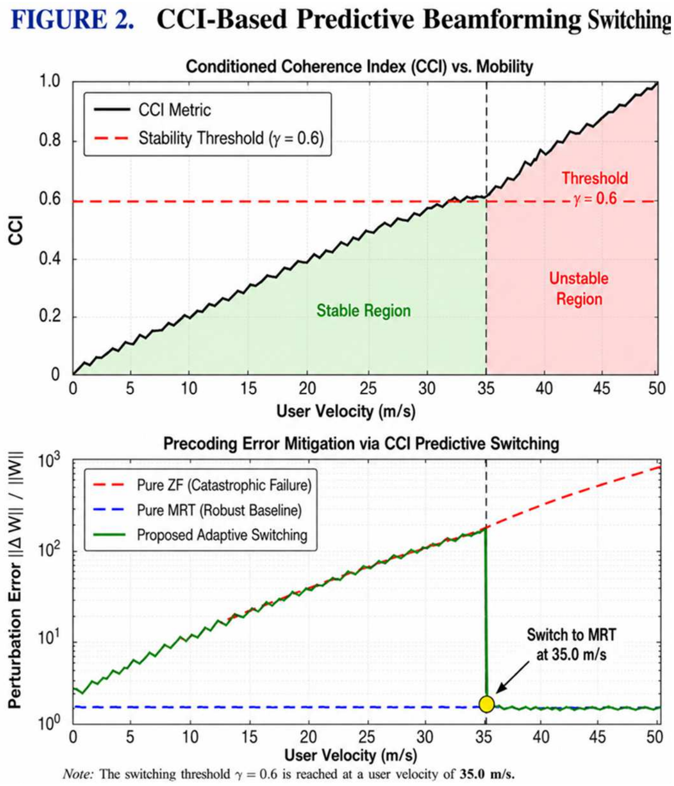

Fig. 2. CCI-based predictive beamforming switching

The top panel shows CCI as a function of mobility, with the instability threshold at $\boldsymbol{\gamma} = 0.6$. The bottom panel shows the perturbation error $\| \Delta W \| / \| W \|$on a logarithmic scale, illustrating rapid ZF degradation beyond this threshold. CCI-based switching transitions to MRT, maintaining stable performance beyond the Coherence Cliff. Perturbation error increases sharply as CCI approaches 0.6, supporting its use as the switching threshold. Together, Figs. 1–2 demonstrate that CCI predicts the Coherence Cliff and enables proactive switching from ZF to MRT.

VII. CONCLUSION

This work introduced the Coherence Cliff as a fundamental phenomenon governing beamforming instability in high-frequency systems and proposed the Conditioned Coherence Index (CCI) as a unified predictive metric. By integrating Doppler-induced decorrelation, signal quality, channel matrix conditioning, and CSI aging, the CCI captures the physical and numerical drivers of abrupt performance degradation. Simulation results confirm that the CCI increases consistently with mobility and that a deterministic transition boundary emerges at $\gamma = 0.6$, validating the CCI as a predictive indicator of the Coherence Cliff. Based on this framework, an adaptive beamforming strategy was proposed that switches between ZF and MRT to maintain robustness. The results demonstrate that proactive CCI-based switching prevents the catastrophic failure of ZF precoders, providing a foundation for reliable communication in Wi-Fi 7 and emerging 6G networks.

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